

CALIBRATION & TEST REPORT

coreDAQ Mk 2 SN0001

PASS · CAL_OK

DETECTOR · FRONT END

InGaAs · LOG
logarithmic, 20 mV/dB nominal

TIER · FIRMWARE

HIGH · HIGHBW
FW v1.6 · $F_{\max}$ 1 MS/s · SYNC

CALIBRATION

1550.0 nm · 2026-09-17
Operator Ravi · 128-pt LUT

IDENTITY

UID 1D00...373530
calibration record verified

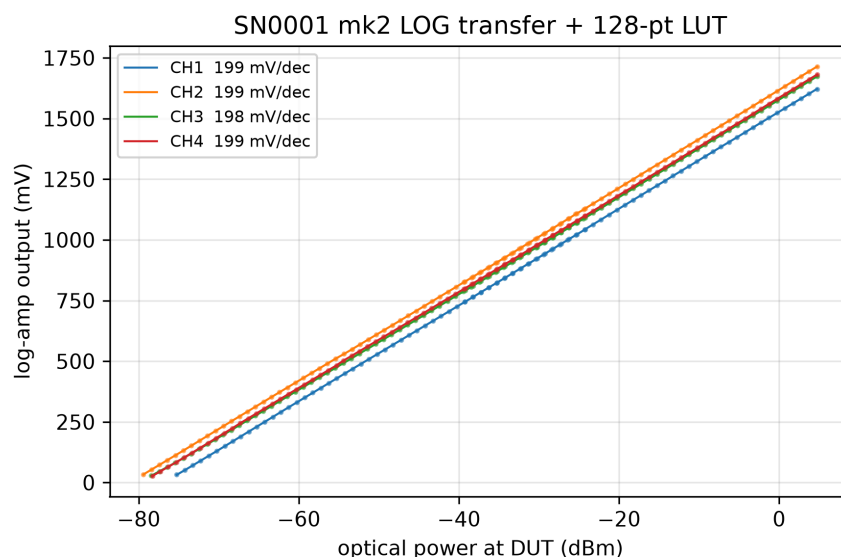


Fig. 1 — Log-amp output vs optical power, CH1-CH4, with 128-pt LUT.

PER-CHANNEL LUT FIT

CH	SLOPE	P_{INT}	R^2	RESID RMS	RESID MAX	DARK	RANGE
1	19.87 mV/dB	20.9 pW	0.999990	6.8 mdB	50.6 mdB	10.4 mV	-75.3 ... +4.7 dBm
2	19.93 mV/dB	8.0 pW	0.999965	6.2 mdB	28.7 mdB	9.3 mV	-79.4 ... +4.7 dBm
3	19.84 mV/dB	11.8 pW	0.999947	7.5 mdB	56.7 mdB	7.4 mV	-78.5 ... +4.8 dBm
4	19.90 mV/dB	11.1 pW	0.999988	9.7 mdB	42.7 mdB	11.8 mV	-78.3 ... +4.8 dBm

METHODOLOGY

A 1550 nm laser is stepped through a calibrated attenuator in 1 dB increments from +5 dBm down to -80 dBm and applied to each channel in turn. Optical power at the unit is referenced to a NIST-traceable power meter; below the meter's floor, power is derived from the attenuator's calibrated transfer. Each point is the average of repeated snapshots after settling. A monotonic 128-point lookup table is fitted per channel and written to the device. Head 27.7 °C / 34.5 %RH during calibration.

STORED IN DEVICE

Monotonic 128-point calibration table per channel, fitted over -80.0 ... +4.8 dBm, written to the unit's non-volatile calibration memory together with a metadata record (serial, date, wavelength, tier, fit summary) and the nominal InGaAs responsivity curve per head (1000 - 1625 nm; 0.990 A/W at 1550 nm).

CALIBRATION & TEST REPORT

coreDAQ Mk 2 SN0002

PASS · CAL_OK

DETECTOR · FRONT END

InGaAs · LOG

logarithmic, 20 mV/dB nominal

TIER · FIRMWARE

HIGH · HIGHBW

FW v1.6 · $F_{\max}$ 1 MS/s · SYNC

CALIBRATION

1550.0 nm · 2026-09-17

Operator Ravi · 128-pt LUT

IDENTITY

UID 3C00...353937

calibration record verified

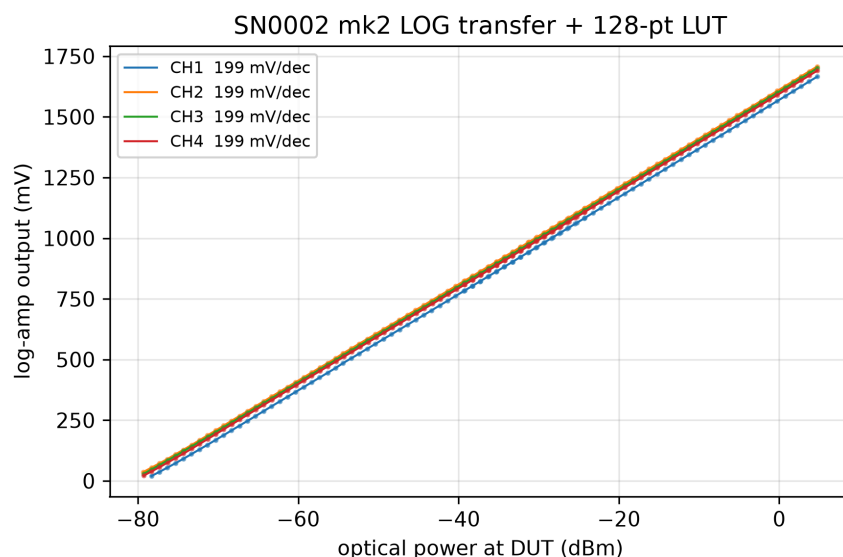


Fig. 1 — Log-amp output vs optical power, CH1-CH4, with 128-pt LUT.

PER-CHANNEL LUT FIT

CH	SLOPE	P_{INT}	R^2	RESID RMS	RESID MAX	DARK	RANGE
1	19.89 mV/dB	13.3 pW	0.999972	5.4 mdB	26.1 mdB	10.1 mV	-78.3 ... +4.7 dBm
2	19.93 mV/dB	8.5 pW	0.999981	5.9 mdB	20.1 mdB	13.6 mV	-79.3 ... +4.7 dBm
3	19.91 mV/dB	9.2 pW	0.999975	5.2 mdB	20.8 mdB	7.0 mV	-79.3 ... +4.7 dBm
4	19.91 mV/dB	10.1 pW	0.999982	7.8 mdB	44.8 mdB	8.9 mV	-79.3 ... +4.7 dBm

METHODOLOGY

A 1550 nm laser is stepped through a calibrated attenuator in 1 dB increments from +5 dBm down to -80 dBm and applied to each channel in turn. Optical power at the unit is referenced to a NIST-traceable power meter; below the meter's floor, power is derived from the attenuator's calibrated transfer. Each point is the average of repeated snapshots after settling. A monotonic 128-point lookup table is fitted per channel and written to the device. Head 24.2 °C / 41.3 %RH during calibration.

STORED IN DEVICE

Monotonic 128-point calibration table per channel, fitted over -80.0 ... +4.8 dBm, written to the unit's non-volatile calibration memory together with a metadata record (serial, date, wavelength, tier, fit summary) and the nominal InGaAs responsivity curve per head (1000 - 1625 nm; 0.990 A/W at 1550 nm). Unit arrived uncalibrated and now reports CAL_OK.

Rise time & bandwidth vs photocurrent

LOW tier vs HIGH tier

Both SN0001 and SN0002 ship as HIGH tier (HIGHBW firmware, $F_{\max}$ 1 MS/s). The logarithmic front end's small-signal bandwidth is set by the photodiode current: at high currents the -3 dB point is fixed by the analog filter, at low currents by the log-amp's input stage. The LOW tier caps the filter at 10 kHz and sampling at 100 kS/s; the HIGH tier opens both to 150 kHz and 1 MS/s.

PHOTOCURRENT	OPTICAL POWER ¹	LOW · BW (-3 DB)	LOW · RISE 10-90 %	HIGH · BW (-3 DB)	HIGH · RISE 10-90 %
1 mA	≈ 1 mW · 0 dBm	10 kHz	35 μs	150 kHz	2.3 μs
100 μA	≈ 100 μW · -10 dBm	10 kHz	35 μs	150 kHz	2.3 μs
10 μA	≈ 10 μW · -20 dBm	10 kHz	35 μs	150 kHz	2.3 μs
1 μA	≈ 1 μW · -30 dBm	10 kHz	35 μs	150 kHz	2.3 μs
100 nA	≈ 100 nW · -40 dBm	10 kHz	35 μs	150 kHz	2.3 μs
10 nA	≈ 10 nW · -50 dBm	10 kHz	35 μs	150 kHz	2.3 μs
1 nA	≈ 1 nW · -60 dBm	10 kHz	35 μs	50 kHz	7 μs
100 pA	≈ 100 pW · -70 dBm	< 10 kHz ²	> 35 μs ²	< 50 kHz ²	> 7 μs ²

¹ InGaAs, 0.99 A/W at 1550 nm. Bandwidth figures from the Mk 2 logarithmic-variant specification; rise time (10-90 %) computed as $0.35 / f_{-3\text{dB}}$ for a single-pole response. Sampling rate limits the resolvable rise time to ≥ 2 samples: 20 μs at 100 kS/s, 2 μs at 1 MS/s. ² Below 1 nA the log-amp input stage dominates; bandwidth falls with photocurrent and is not specified.

DELIVERED CONFIGURATION

SN0001	HIGH · HIGHBW · 1 MS/s · SYNC	SN0002	HIGH · HIGHBW · 1 MS/s · SYNC
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Tier lock reads LOCK=MATCH on both units; the license is bound to the chip UID recorded on pages 1 and 2.